

ASS 17.5

AI ASSISTED CODING

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Import libraries

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler, LabelEncoder

Load dataset

data = pd.read_csv("loanapproval.csv")

Preview data

print("Initial Data:")

print(data.head())

print("\nMissing Values:\n", data.isnull().sum())

1. Handle Missing Values

Fill numerical columns with median (robust choice)

data['loan_amount'].fillna(data['loan_amount'].median(), inplace=True)

data['credit_score'].fillna(data['credit_score'].median(), inplace=True)

2. Encode Target Variable

Convert loan_status (Approved/Rejected) → (1/0)

le = LabelEncoder()

data['loan_status'] = le.fit_transform(data['loan_status'])

```

# -----

# 3. Separate Features & Target
# -----

X = data.drop('loan_status', axis=1)
y = data['loan_status']


# -----

# 4. Handle Categorical Features (if any)
# -----

# Convert categorical columns into dummy variables
X = pd.get_dummies(X, drop_first=True)


# -----

# 5. Scale Numerical Features
# -----

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)


# -----

# 6. Train-Test Split
# -----

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y,
    test_size=0.2,
    random_state=42
)

```

```
# -----
```

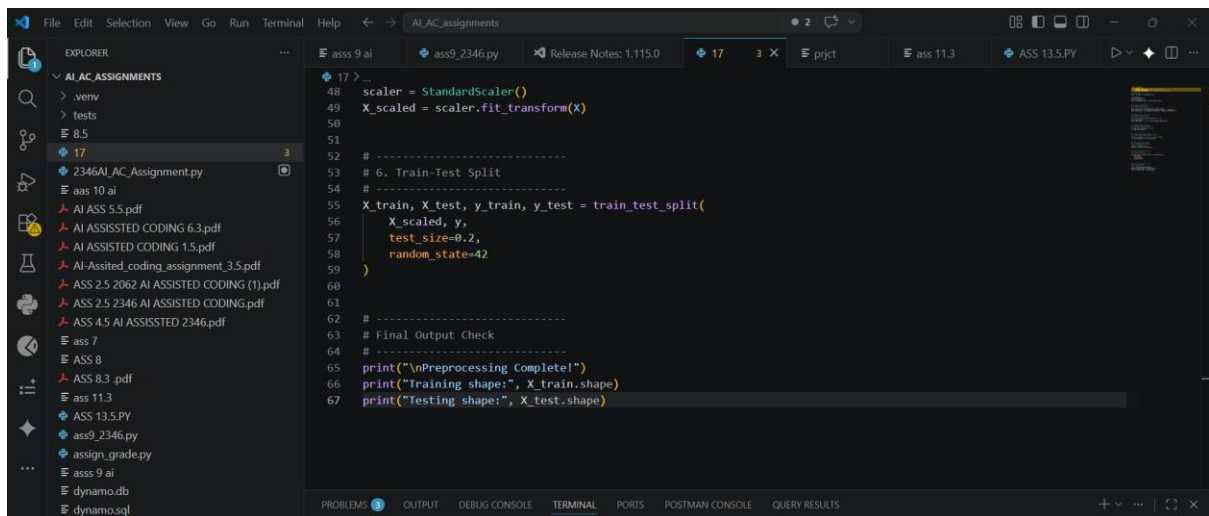
```
# Final Output Check
```

```
# -----
```

```
print("\nPreprocessing Complete!")
```

```
print("Training shape:", X_train.shape)
```

```
print("Testing shape:", X_test.shape)
```



Social Media Profiles

Preprocess a social media engagement dataset using AI assistance.

Instructions:

- Remove duplicate user profiles.
- Handle missing values
- Encode categorical columns

Dataset link:

<https://www.kaggle.com/datasets/medaxone/top-100-social-media-profiles>

Expected Output:

- A cleaned dataset for engagement prediction.

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99 for col in num_cols:
100     data[col].fillna(data[col].median(), inplace=True)
101
102 # Fill categorical columns with mode
103 cat_cols = data.select_dtypes(include=['object']).columns
104 for col in cat_cols:
105     data[col].fillna(data[col].mode()[0], inplace=True)
106
107 print("\nMissing values after cleaning:\n", data.isnull().sum())
108
109 # -----
110 # 3. Encode Categorical Columns
111 # -----
112 # Option 1: Label Encoding (for simplicity)
113 le = LabelEncoder()
114
115 for col in cat_cols:
116     data[col] = le.fit_transform(data[col])
117
118 # -----
119 # Final Clean Dataset
120 # -----
121
122 print("\nCleaned Data Sample:")
123 print(data.head())
```

```
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AI_AC_assignments
ass9_2346.py Release Notes: 1.115.0 17 8 prjct ass 11.3 ASS 13.5.PY
EXPLORER
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186 data = pd.get_dummies(data, columns=['season'], drop_first=True)
187
188 # -----
189 # 5. Normalize Rainfall
190 # -----
191 scaler = MinMaxScaler()
192 data['rainfall'] = scaler.fit_transform(data[['rainfall']])
193
194 # -----
195 # Final Output
196 # -----
197
198 print("\nProcessed Data Sample:")
199 print(data.head())
200
201 # Save processed dataset
202 data.to_csv("processed_weather_data.csv", index=False)
203
204
205 print("\n✅ Weather dataset is ready for forecasting!")
```

Movie Ratings Dataset Preparation

Task:

Clean and preprocess a movie ratings dataset using AI-generated scripts.

Instructions:

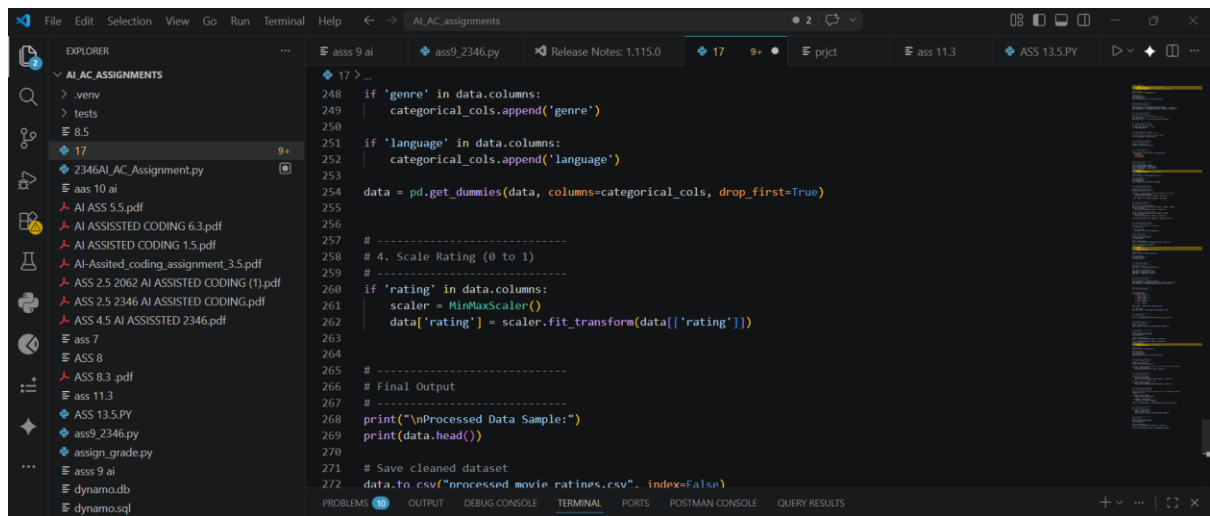
- Handle missing values in rating, genre.
- Encode categorical variables like genre, language.
- Remove duplicate movie entries.
- Scale rating values between 0 and 1.

Dataset link:

<https://www.kaggle.com/datasets/PromptCloudHQ/imdb-data>

Expected Output:

- A dataset suitable for recommendation systems.



The screenshot shows a Visual Studio Code editor window with a file explorer on the left and a code editor in the center. The file explorer shows a project named 'AI.AC_ASSIGNMENTS' with various files including PDFs, Python scripts, and a database. The code editor displays a Python script named 'ass9_2346.py' with the following code:

```
248 if 'genre' in data.columns:
249     categorical_cols.append('genre')
250
251 if 'language' in data.columns:
252     categorical_cols.append('language')
253
254 data = pd.get_dummies(data, columns=categorical_cols, drop_first=True)
255
256 # -----
257 # 4. Scale Rating (0 to 1)
258 # -----
259 if 'rating' in data.columns:
260     scaler = MinMaxScaler()
261     data['rating'] = scaler.fit_transform(data[['rating']])
262
263 # -----
264 # Final Output
265 # -----
266 print("\nProcessed Data Sample:")
267 print(data.head())
268
269 # Save cleaned dataset
270 data.to_csv("processed movie ratings.csv", index=False)
```